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## PUNCHING DEVICE AND PUNCHING METHOD USING SAME

### Abstract


A punching device includes: a die to receive a substrate disposed thereon; a stripper spaced apart from and facing the die, and to move toward the substrate to fix the substrate; and a punch to move toward the substrate and cut the substrate. The stripper includes: a punch accommodation portion to accommodate the punch; and a punching oil supply passage located inside the stripper, and to supply a punching oil to a side surface of the punch accommodation portion.



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### Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and the benefit of Korean Application No. 10-2024-0032255, filed on Mar. 6, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference herein.

BACKGROUND

1. Field

[0002] Aspects of embodiments of the present disclosure relate to a punching device, more particularly, including a punching oil supply passage positioned inside a stripper, and a punching method using the punching device.

2. Description Of The Related Art

[0003] Unlike primary batteries that are not designed to be (re) charged, secondary (or rechargeable) batteries are batteries that are designed to be discharged and recharged. Low-capacity secondary batteries are used in portable, small electronic devices, such as smart phones, feature phones, notebook computers, digital cameras, and camcorders, while large-capacity secondary batteries are widely used as power sources for driving motors in hybrid vehicles and electric vehicles and for storing power (e.g., home and/or utility scale power storage). A secondary battery generally includes an electrode assembly composed of a positive electrode and a negative electrode, a case accommodating the same, and electrode terminals connected to the electrode assembly.

[0004] A positive electrode plate and a negative electrode plate of a lithium-ion battery may be manufactured by applying a cathode active material to a positive electrode current collector in the form of a metal thin-film (e.g., an aluminum foil), and applying an anode active material to a negative electrode current collector in the form of a metal thin-film (e.g., a copper foil). The electrode plate may be manufactured through mixing, coating, and pressing processes to manufacture a roll-shaped metal thin-film substrate coated with an active material, and slitting and notching processes to cut the roll-shaped metal thin-film substrate in accordance with desired battery specifications.

[0005] The above information disclosed in this Background section is for enhancement of understanding of the background of the present disclosure, and therefore, it may contain information that does not constitute related (or prior) art.

SUMMARY

[0006] An electrode plate that satisfies desired specifications may be manufactured by blanking a substrate using a punching device through a notching process. Punching oil is sprayed on the substrate, such that the punching oil is applied to the substrate, and blanking is performed on the substrate to which the punching oil has been applied. In a case where the specifications of the electrode plate change, it may be desirable to install a new punching oil spraying device or change settings in accordance with the specifications of the electrode plate. However, it may take a lot of time and costs to install a new punching oil spraying device or to change the settings.

[0007] Embodiments of the present disclosure may be directed to a punching device, and a punching method using the punching device.

[0008] These and other aspects and features of the present disclosure will be described in or will be apparent from the following description of embodiments of the present disclosure.

[0009] According to one or more embodiments of the present disclosure, a punching device includes: a die configured to receive a substrate disposed thereon; a stripper spaced apart from and facing the die, and configured to move toward the substrate to fix the substrate; and a punch configured to move toward the substrate and cut the substrate. The stripper includes: a punch accommodation portion configured to accommodate the punch; and a punching oil supply passage located inside the stripper, and configured to supply a punching oil to a side surface of the punch accommodation portion.

[0010] In an embodiment, the punch accommodation portion may have a shape of a through-hole through which the punch may be configured to be inserted via the stripper.
[0011] In an embodiment, one side surface of the punch accommodation portion may be opened.
[0012] In an embodiment, the punch accommodation portion may include a plurality of side surfaces, a nozzle of the punching oil supply passage may be located on a first side surface from among the plurality of side surfaces of the punch accommodation portion, and a side surface of the punch may be configured to face the first side surface.
[0013] In an embodiment, the punching oil supply passage may include a plurality of punching oil supply passages, a first nozzle of a first punching oil supply passage and a second nozzle of a second punching oil supply passage from among the plurality of punching oil supply passages may be located on the first side surface of the punch accommodation portion, and the side surface of the punch may be configured to face the first side surface.
[0014] In an embodiment, the punching oil supply passage may include a plurality of punching oil supply passages, a first nozzle of a first punching oil supply passage from among the plurality of punching oil supply passages may be located on the first side surface of the punch accommodation portion, a second nozzle of a second punching oil supply passage from among the plurality of punching oil supply passages may be located on a second side surface from among the plurality of side surfaces of the punch accommodation portion, and the side surface of the punch may be configured to face the first side surface.
[0015] In an embodiment, the stripper may further include an air supply passage located inside the stripper, and configured to supply air to the side surface of the punch accommodation portion.
[0016] In an embodiment, the punch may include an air supply hole extending through the punch, and configured to supply air to a bottom surface of the punch.
[0017] In an embodiment, the air may be supplied at a pressure of 0.1 Mpa or more.
[0018] In an embodiment, the air supply passage may be located below the punching oil supply passage.
[0019] In an embodiment, the stripper may include a plurality of punch accommodation portions configured to accommodate a plurality of punches, and a plurality of punching oil supply passages configured to supply the punching oil. A first nozzle of a first punching oil supply passage from among the plurality of punching oil supply passages may be located on a side surface of a first punch accommodation portion from among the plurality of punch accommodation portions, and a second nozzle of a second punching oil supply passage from among the plurality of punching oil supply passages may be located on a side surface of a second punch accommodation portion from among the plurality of punch accommodation portions. A side surface of a first punch may be configured to face the side surface of the first punch accommodation portion, and a side surface of a second punch may be configured to face the side surface of the second punch accommodation portion.
[0020] In an embodiment, the punching oil may be supplied in a fixed-quantity discharge method.
[0021] In an embodiment, the punching oil may be supplied at a predetermined flow rate.
[0022] In an embodiment, the punching device may further include a controller configured to adjust an amount of the punching oil. The controller may be configured to control a pump connected to the punching oil supply passage to supply the punching oil in proportion to a number of times the punch is raised and lowered.
[0023] In an embodiment, the punching device may further include a controller configured to adjust an amount of the air. The controller may be configured to control an air pump connected to the air supply passage to supply the air in proportion to a number of times the punch is raised and lowered.
[0024] According to one or more embodiments of the present disclosure, a punching method includes: supplying a substrate between a stripper and a die; supplying a punching oil to a side surface of a punch accommodation portion of the stripper through a punching oil supply passage

located inside the stripper; and moving a punch and the stripper toward the substrate, such that the stripper fixes the substrate and the punch cuts the substrate. A surface of the punch is disposed to face a side surface of the stripper.

[0025] In an embodiment, the supplying of the punching oil to the side surface of the punch accommodation portion may include supplying the punching oil to a plurality of punching oil supply passages by a pump, such that the punching oil may be sequentially supplied through a plurality of nozzles located on the side surface of the punch accommodation portion.

[0026] In an embodiment, the method may further include removing foreign materials attached to the surface of the punch by supplying air to the surface of the punch using at least one of an air supply passage located in the stripper or an air supply hole located in the punch.

[0027] In an embodiment, the supplying of the punching oil to the side surface of the punch accommodation portion may include supplying the punching oil to the side surface of the punch accommodation portion before cutting the substrate.

[0028] In an embodiment, the removing of the foreign materials attached to the surface of the punch may include removing the foreign materials attached to the surface of the punch by supplying air to the surface of the punch after punching the substrate.

[0029] According to some embodiments of the present disclosure, even in a case where friction occurs when the punch and the metal thin-film substrate come into contact with each other, damage to the punch may be prevented or substantially prevented.

[0030] According to some embodiments of the present disclosure, repeated punching may prevent or substantially prevent foreign materials (e.g., a metal powder and the like) from being stuck to the mold, or may prevent or substantially prevent burrs from being formed on the substrate.

[0031] According to some embodiments of the present disclosure, the sticking of the foreign materials and the formation of the burrs may be effectively prevented by supplying the punching oil to the side surface of the punch and the side surface of the punch accommodation portion of the stripper.

[0032] According to some embodiments of the present disclosure, low voltage defects and short circuit defects in secondary batteries may be prevented or substantially prevented.

[0033] According to some embodiments of the present disclosure, the removal of the foreign materials stuck/attached to the surface of the punch may be further facilitated by the air supply nozzle and/or the air supply hole.

[0034] According to some embodiments of the present disclosure, the punching device may supply the punching oil directly to the cut region, such that the punching oil may be supplied evenly or substantially evenly.

[0035] According to some embodiments of the present disclosure, because the stripper may not be designed to directly cut the metal thin-film substrate, the corresponding passage or nozzle may not be easily damaged, even in a case where the stripper includes a punching oil supply passage, a punching oil supply nozzle, an air supply passage, and/or an air supply nozzle, thereby improving the durability of the punching device.

[0036] According to some embodiments of the present disclosure, the air supply nozzle may be disposed to face the side surface of the punch, such that air with a stronger pressure may be supplied to the side surface of the punch in which a relatively larger amount of powder may be attached.

[0037] According to some embodiments of the present disclosure, the air supply nozzle may prevent or substantially prevent flowing punching oil from flowing into the air supply nozzle by supplying air to the side surface of the punch accommodation portion.

[0038] According to some embodiments of the present disclosure, in the structure in which the plurality of punches are raised and lowered, the punching oil may be effectively supplied to the plurality of punches by forming a passage with a simpler structure in (e.g., only in) the stripper, without forming the passage in each punch.

[0039] According to some embodiments of the present disclosure, the punching oil may be supplied more smoothly by forming the punching oil supply nozzles on the plurality of side surfaces of the stripper.
[0040] According to some embodiments of the present disclosure, the punching device may collectively control the supply of the punching oil through the punching oil passage, making it easier to manage the supply of the punching oil.
[0041] However, aspects and features of the present disclosure are not limited to those described above, and other aspects and features not mentioned will be clearly understood by a person skilled in the art from the detailed description, described below.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0042] The following drawings attached to this specification illustrate embodiments of the present disclosure, and further describe aspects and features of the present disclosure together with the detailed description of the present disclosure. Thus, the present disclosure should not be construed as being limited to the drawings:
[0043] FIG. **1** illustrates an exploded perspective view of a punching device according to an embodiment of the present disclosure;
[0044] FIG. **2** illustrates a front view of a punching device according to an embodiment of the present disclosure;
[0045] FIG. **3** illustrates a side view of a punching device according to an embodiment of the present disclosure;
[0046] FIG. **4** illustrates a side view of a punching device according to an embodiment of the present disclosure;
[0047] FIG. **5** illustrates a side view showing a state in which a punching oil is supplied in a punching device according to an embodiment of the present disclosure;
[0048] FIG. **6** illustrates a side view showing a state in which a metal thin-film substrate is punched by a punching device according to an embodiment of the present disclosure;
[0049] FIG. **7** illustrates a side view showing a state in which air is supplied in a punching device according to an embodiment of the present disclosure;
[0050] FIG. **8** illustrates a perspective view of a stripper according to an embodiment of the present disclosure;
[0051] FIG. **9** illustrates a perspective view of a stripper according to an embodiment of the present disclosure; and
[0052] FIG. **10** illustrates a flowchart showing an example of a punching method according to an embodiment of the present disclosure.
DETAILED DESCRIPTION
[0053] Hereinafter, embodiments of the present disclosure will be described, in detail, with reference to the accompanying drawings. The terms or words used in the present specification and claims are not to be limitedly interpreted as general or dictionary meanings and should be interpreted as meanings and concepts that are consistent with the technical idea of the present disclosure on the basis of the principle that an inventor can be his/her own lexicographer to appropriately define concepts of terms to describe his/her invention in the best way.
[0054] The embodiments described in this specification and the configurations shown in the drawings are only some of the embodiments of the present disclosure and do not represent all of the technical spirit, aspects, and features of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that can replace or modify the embodiments described herein at the time of filing this application.

[0055] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected, or coupled to the other element or layer or one or more intervening elements or layers may also be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements.
[0056] In the figures, dimensions of the various elements, layers, etc. may be exaggerated for clarity of illustration. The same reference numerals designate the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” Expressions, such as “at least one of” and “any one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. When phrases such as “at least one of A, B and C, “at least one of A, B or C,” “at least one selected from a group of A, B and C,” or “at least one selected from among A, B and C” are used to designate a list of elements A, B and C, the phrase may refer to any and all suitable combinations or a subset of A, B and C, such as A, B, C, A and B, A and C, B and C, or A and B and C. As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.
[0057] It will be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.
[0058] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.
[0059] The terminology used herein is for the purpose of describing embodiments of the present disclosure and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
[0060] Also, any numerical range disclosed and/or recited herein is intended to include all sub-ranges of the same numerical precision subsumed within the recited range. For example, a range of

"1.0 to 10.0" is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein. All such ranges are intended to be inherently described in this specification such that amending to expressly recite any such subranges would comply with the requirements of 35 U.S.C. § 112(a) and 35 U.S.C. § 132(a).
[0061] References to two compared elements, features, etc. as being "the same" may mean that they are "substantially the same". Thus, the phrase "substantially the same" may include a case having a deviation that is considered low in the art, for example, a deviation of 5% or less. In addition, when a certain parameter is referred to as being uniform in a given region, it may mean that it is uniform in terms of an average.
[0062] Throughout the specification, unless otherwise stated, each element may be singular or plural.
[0063] Arranging an arbitrary element "above (or below)" or "on (under)" another element may mean that the arbitrary element may be disposed in contact with the upper (or lower) surface of the element, and another element may also be interposed between the element and the arbitrary element disposed on (or under) the element.
[0064] In addition, it will be understood that when a component is referred to as being "linked," "coupled," or "connected" to another component, the elements may be directly "coupled," "linked" or "connected" to each other, or another component may be "interposed" between the components".
[0065] Throughout the specification, when "A and/or B" is stated, it means A, B or A and B, unless otherwise stated. That is, "and/or" includes any or all combinations of a plurality of items enumerated. When "C to D" is stated, it means C or more and D or less, unless otherwise specified.
[0066] FIG. **1** illustrates an exploded perspective view of a punching device **100** according to an embodiment of the present disclosure.
[0067] The punching device **100** according to an embodiment of the present disclosure may include (e.g., may be) a punching mold that may punch a continuously provided metal thin-film substrate, and cut the thin-film substrate into substrates satisfying a desired standard (e.g., a predetermined standard). Referring to FIG. **1**, the punching device **100** may include a first mold **110** and a second mold **120**. The first mold **110** may refer to an upper mold. The second mold **120** may refer to a lower mold.
[0068] As used herein, an upper portion may be a direction toward the first mold **110** (or a portion thereof), and a lower portion may be a direction toward the second mold **120** (or a portion thereof). In an embodiment, a substrate coated with an active material may be provided to the punching device **100**.
[0069] The first mold **110** may include a first base plate **112**, a stripper plate **114**, a punch **116**, and a stripper **118**. The stripper plate **114** may be fixedly connected to (e.g., coupled to or attached to) the lower portion of the first base plate **112**. The stripper **118** may be connected to (e.g., coupled to or attached to) the stripper plate **114**. For example, a stripper connection pin may connect the stripper **118** to the stripper plate **114**. The stripper **118** may include a punch accommodation portion **118*a***. The punch **116** may be connected to (e.g., coupled to or attached to) the stripper plate **114**, and may be accommodated in the punch accommodation portion **118*a*** of the stripper **118**.
[0070] Referring to FIG. **1**, one side of the punch accommodation portion **118*a*** is shown as an open structure. However, the present disclosure is not limited thereto, and the punch accommodation portion **118*a*** may be in the form of a through-hole through which the punch **116** is inserted via the

stripper **118**. Some embodiments of the punch accommodation portion **118***a* are described in more detail below with reference to FIGS. **8** and **9**.

[0071] The stripper **118** may include a punching oil supply passage **118***b* positioned inside the stripper **118**. The punching oil supply passage **118***b* may supply punching oil to the side surface of the punch accommodation portion **118***a*. In some embodiments, the side surface of the punch **116** may be disposed to face the side surface of the punch accommodation portion **118***a*, such that punching oil may be supplied to the side surface of the punch **116**.

[0072] In an embodiment, the punching oil supply passage **118***b* may be positioned inside the stripper **118**, and the stripper **118** may further include an air supply passage through which air is supplied to the side surface of the punch accommodation portion **118***a*. In some embodiments, the air supply passage may be positioned below the punching oil supply passage **118***b*. In some embodiments, the side surface of the punch **116** may be disposed to face the side surface of the punch accommodation portion **118***a*, such that air may be supplied to the side surface of the punch **116**.

[0073] The second mold **120** may include a second base plate **124** and a die **122**. The die **122** may be fixedly connected to (e.g., coupled to or attached to) the second base plate **124**. The die **122** may be fixedly connected to (e.g., coupled to or attached to) the upper portion of the second base plate **124**. The die **122** may include a punch insertion portion **122***a*.

[0074] The first mold **110** and the second mold **120** may be spaced apart from each other, and may be connected to each other. For example, a base guide post may connect the first base plate **112** to the second base plate **124**, such that the first mold **110** and the second mold **120** are connected to (e.g., coupled to or attached to) each other. However, the present disclosure is not limited thereto, and the punching device according to embodiments of the present disclosure may further include other desired components in the coupling structure between the first mold **110** and the second mold **120** for cutting the metal thin-film substrate.

[0075] The first mold **110** may descend or ascend toward the second mold **120**. In some embodiments, the second mold **120** may descend or ascend toward the first mold **110**. In a case where the first mold **110** descends, the stripper **118** may come into contact with the die **122**. In a case where the metal thin-film substrate is disposed on the die **122**, the stripper **118** may be fixed to be in contact with the substrate. In some embodiments, in a case where the first mold **110** descends, the punch **116** may descend and be inserted into the punch insertion portion **122***a*. In a case where the metal thin-film substrate is disposed on the die **122**, the punch **116** may punch and cut the metal thin-film substrate. The cut metal thin-film substrate may be separated through the punch insertion portion **122***a*.

[0076] The punch insertion portion **112***a* and/or the punch **116** may be formed in a shape corresponding to a desired standard (e.g., a predetermined standard). For example, the punch insertion portion **112***a* and/or the punch **116** may correspond to the cut shape of the metal thin-film substrate. Similarly, the punch accommodation portion **118***a* may be formed to correspond to the shape of the punch **116**, so as to accommodate the punch **116**.

[0077] In addition to the structure and the configuration described above, the punching device **100** illustrated in FIG. **1** may further include a configuration and components desired for manufacturing an electrode plate for a secondary battery as would be understood by those having ordinary skill in the art.

[0078] FIG. **2** illustrates a front view of a punching device **200** according to an embodiment of the present disclosure.

[0079] The punching device **200** according to an embodiment of the present disclosure may manufacture an electrode plate for a secondary battery of a desired standard (e.g., a certain or predetermined standard) by lowering the first mold **210** to fix the metal thin-film substrate **230**, and cutting a portion of the metal thin-film substrate **230**. The punching device **200** may include a structure in which the punching device **100** illustrated in FIG. **1** is assembled.

[0080] The metal thin-film substrate **230** may be provided between the first mold **210** and the second mold **220**. The metal thin-film substrate **230** may be disposed on the die **222**. In some embodiments, the disposed metal thin-film substrate **230** may be a cutting target. Thereafter, as the first mold **210** descends, the first mold **210** and the second mold **220** may fix the metal thin-film substrate **230** therebetween. In some embodiments, as the stripper plate **212** descends, the punch **214** and the stripper **216** connected to (e.g., coupled to or attached to) the stripper plate **212** may descend together. In some embodiments, the second base plate **224** connected to (e.g., coupled to or attached to) the die **222** may be fixed. The stripper **216** may fix the metal thin-film substrate **230** disposed on the die **222**.
[0081] The punch **214** may be lowered together with the stripper plate **212**, and the metal thin-film substrate **230** may be cut to a desired standard (e.g., a predetermined standard). For example, while the first mold **210** is lowered and the metal thin-film substrate **230** is fixed between the stripper **216** and the die **222**, the punch **214** may further descend. The stripper **216** may be connected to the stripper plate **212** by using an elastic connection member (e.g., a spring pin). As such, the stripper plate **212** may further descend in a state where the stripper **216** fixes the metal thin-film substrate **230**. The punch **214**, which additionally descends together with the stripper plate **212**, may be inserted into the punch insertion portion **222***a*. In the process of inserting the punch **214** into the punch insertion portion **222***a*, the metal thin-film substrate **230** may be cut to a desired standard (e.g., a predetermined standard).
[0082] In some embodiments, the surface of the punch **214** may be in a state of being supplied with a punching oil. The surface of the punch **214** may include a side surface of the punch **214** and a bottom surface of the punch **214**. For example, the punching oil supplied through the punching oil supply passage included in the stripper **216** may be supplied to the surface of the punch **214**. As such, even in a case where friction occurs when the punch **214** and the metal thin-film substrate **230** come into contact with each other, damage to the punch **214** may be prevented or substantially prevented.
[0083] In some embodiments, as punching is repeated, foreign materials (e.g., metal powder and the like) may stick to the molds **210** and **220**, or burrs may be formed therein. In the punching device **200** according to an embodiment of the present disclosure, the punching oil may be supplied to the side surface of the punch **214** and the side surface of the punch accommodation portion of the stripper **216** (e.g., the punch accommodation portion **118***a* in FIG. **1**). The supplied punching oil may prevent or substantially prevent the sticking of the foreign materials and the formation of the burrs. Low voltage defects, short circuit defects, and the like of secondary batteries manufactured by using the punching device **200** according to one or more embodiments of the present disclosure may be prevented or substantially prevented.
[0084] FIG. **3** illustrates a side view of a punching device according to an embodiment of the present disclosure.
[0085] The punching device may include a stripper **310**, a punch **320**, and a die **330**. The stripper **310** may include a punching oil supply passage **312** and an air supply passage **316** therein. In some embodiments, the air supply passage **316** may be positioned below the punching oil supply passage **312**. In some embodiments, the stripper **310** may include a punch accommodation portion in which the punch **320** may be accommodated.
[0086] Referring to FIG. **3**, as the first mold (e.g., the first mold **110** in FIG. **1** and the first mold **210** in FIG. **2**) descends, the metal thin-film substrate **340** may be in a state of being fixed by the die **330** and the stripper **310**. As the first mold further descends, the punch **320** may punch and cut the metal thin-film substrate **340**. The process by which the punch **320** cuts the metal thin-film substrate **340** will be described in more detail below with reference to FIGS. **5** to **7**.
[0087] In an embodiment, a punching oil supply nozzle **314** corresponding to a punching oil outlet may be formed on the side surface of the punch accommodation portion. The punching oil supply passage **312** may supply a punching oil through the punching oil supply nozzle **314**, such that the

punching oil may be supplied to the side surface of the punch accommodation portion and/or the side surface of the punch **320**. In some embodiments, an air supply nozzle **318** corresponding to an air outlet may be formed on the side surface of the punch accommodation portion. The air supply passage **316** may supply air through the air supply nozzle **318**, such that the air may be supplied to the side surface of the punch accommodation portion and/or the side surface of the punch **320**.

[0088] In an embodiment, the punching device may further include a pump **350** and an air pump **360**. The pump **350** may be connected to the punching oil supply passage **312**. The pump **350** may supply the punching oil to the punching oil supply passage **312**. The punching oil may be supplied to the side surface of the punch accommodation portion and/or the side surface of the punch **320** through the punching oil supply passage **312** and the punching oil supply nozzle **314**. The air pump **360** may be connected to the air supply passage **316**. The air pump **360** may supply air to the air supply passage **316**. The air may be supplied to the side surface of the punch accommodation portion and/or the side surface of the punch **320** through the air supply passage **316** and the air supply nozzle **318**.

[0089] In an embodiment, the pump **350** may supply the punching oil at a suitable flow rate (e.g., a predetermined flow rate). For example, the pump **350** may continuously supply the punching oil to the punching oil supply passage **312** at the flow rate (e.g., the predetermined flow rate), regardless of the number of times the punch **320** is raised and lowered. In other embodiments, the pump **350** may supply the punching oil in a fixed-quantity discharge method. The fixed-quantity discharge method may refer to a method of repeatedly discharging an amount (e.g., a certain amount) of the punching oil. For example, the pump **350** may correspond to a fixed-quantity discharge pump, and the punching oil may be supplied in a fixed-quantity discharge method. In some embodiments, the pump **350** may supply the punching oil in proportion to the number of times of raising and lowering. The pump **350** may supply the punching oil to the punching oil supply passage **312** at an n-time of the raising/lowering timing, where n is a natural number.

[0090] In an embodiment, the punching device may further include a controller **370**. The controller **370** may control the pump **350** to adjust the amount of punching oil supplied. For example, the controller **370** may adjust the flow rate of the punching oil supplied to the punching oil supply passage **312**. In some embodiments, the controller **370** may adjust an amount (e.g., a certain or predetermined amount) of punching oil supplied in a fixed-quantity discharge method. In some embodiments, the controller **370** may control the air pump **360** to adjust the amount of air supplied. For example, the controller **370** may adjust the volume and the atmospheric pressure of the air supplied to the air supply passage **316**. The controller **370** may communicate with the pump **350** and/or the air pump **360** by using various suitable wired or wireless methods.

[0091] In an embodiment, the controller **370** may control the pump **350** to supply the punching oil so as to be associated with the raising and lowering of the punch **320**. For example, the controller **370** may control the pump **350** to supply the punching oil in proportion to the number of times the punch **320** is raised and lowered. For example, the controller **370** may control the pump **350** to supply the punching oil every time the punch **320** is raised and lowered.

[0092] Similarly, the controller **370** may control the air pump **360** to supply the air in association with the raising and lowering of the punch **320**. For example, the controller **370** may control the air pump **360** to supply the air in proportion to the number of times the punch **320** is raised and lowered. For example, the controller **370** may control the air pump **360** to supply the air whenever the punch **320** is raised and lowered.

[0093] FIG. **4** illustrates a side view of a punching device according to an embodiment of the present disclosure.

[0094] Referring to FIG. **4**, a metal thin-film substrate **440** may be fixed by a die **430** and a stripper **410**. The stripper **410** may include a punching oil supply passage **412**.

[0095] In an embodiment, the punch **420** may include an air supply hole **422**, instead of the stripper **410** including an air supply passage (e.g., the air supply passage **316** in FIG. **3**). In some embodiments, the air supply hole **422** may be formed to pass through the punch **420**. The air supply hole **422** may supply air to the bottom surface of the punch **420**.

[0096] The air supply hole **422** may include an air supply line **422***a* and a push pin **422***b*. The push pin **422***b* may include a push pin spring **422***b*_**1** and a push pin hole **422***b*_**2**. The push pin spring **422***b*_**1** assists to lift the descending push pin **422***b* to an original position thereof. The push pin hole **422***b*_**2** may be formed to pass through the inside of the push pin **422***b*.

[0097] The air supply line **422***a* may supply air to the air supply hole **422**. The push pin **422***b* may block air supplied below a certain atmospheric pressure from the air supply line **422***a*. In a case where air supplied above a certain atmospheric pressure is supplied through the air supply line **422***a*, the air may be supplied to the inner space of the air supply hole **422**. The supplied air may be supplied to the bottom surface of the punch **420** through the push pin hole **422***b*_**2**. Thereafter, in a case where the air supplied from the air supply line **422***a* is less than a certain atmospheric pressure, the push pin spring **422***b*_**1** may lift the push pin **422***b* to the original position thereof.

[0098] FIG. **4** illustrates that the punch **420** includes the air supply hole **422**, instead of the stripper **410** including the air supply passage, but the present disclosure is not limited thereto. For example, the stripper **410** may also include the air supply passage, while the punch **420** may include the air supply hole **422**. In this case, the air may be supplied more smoothly to the surface of the punch **420**. In some embodiments, the air supply passage may supply the air to the side surface of the punch **420**, and the air supply hole **422** may supply the air to the bottom surface of the punch **420**. As such, the removal of the foreign materials stuck/attached to the surface of the punch **420** may be made easier.

[0099] FIG. **5** illustrates a side view showing a state in which a punching oil O is supplied in a punching device according to an embodiment of the present disclosure. The punching device described in more detail below with reference to FIGS. **5** to **7** is illustrated as including an air supply passage **515** without an air supply hole, but the present disclosure is not limited thereto. For example, the punching device may include the air supply hole **422** of FIG. **4**.

[0100] The processes described in more detail below with reference to FIGS. **5** to **7** may be performed in the order illustrated in FIGS. **5** to **7**. FIG. **5** may illustrate a state before a punch **520** descends and cuts a metal thin-film substrate **540**. In some embodiments, FIG. **5** illustrates that the metal thin-film substrate **540** is fixed by a die **530** and a stripper **510**, but the present disclosure is not limited thereto. For example, the process of supplying the punching oil O may be performed before a first mold descends and the stripper **510** fixes the metal thin-film substrate **540** disposed on the die **530**. In other embodiments, the process of supplying the punching oil O may be performed regardless of whether the first mold is raised or lowered.

[0101] The punching oil supply passage **512** may supply the punching oil O to a side surface of a punch accommodation portion. In some embodiments, the punching oil O may be supplied through a punching oil supply nozzle **514** formed on the side surface of the punch accommodation portion. In some embodiments, the side surface of the punch **520** may be disposed to face the side surface of the punch accommodation portion in which the punching oil supply nozzle **514** is formed. The supplied punching oil O may flow between the side surface of the punch accommodation portion and the side surface of the punch **520**. In this manner, the punching oil O may be supplied to the side surface of the punch **520**. In some embodiments, the punching oil O may flow between the side surface of the punch accommodation portion and the side surface of the punch **520**, and may be supplied to the cut portion of the metal thin-film substrate **540**.

[0102] FIG. **6** illustrates a side view showing a state in which a metal thin-film substrate **640** is punched by a punching device according to an embodiment of the present disclosure.

[0103] In a state in which the metal thin-film substrate **640** is fixed by a die **630** and a stripper **610**, a punch **620** may be lowered to cut the metal thin-film substrate **640**.

[0104] The metal thin-film substrate **640** may be separated into a cut substrate **642** and a separated substrate **644**. The lowered punch **620** may be raised again after cutting the metal thin-film substrate **640**.

[0105] In the process of lowering the punch **620** to cut the metal thin-film substrate **640**, foreign materials B may be stuck or attached to the surface of the punch **620**. For example, powder generated while cutting the metal thin-film substrate **640** may be stuck or attached to the surface of the punch **620**. In some embodiments, the supplied punching oil may be unevenly stuck to the surface of the punch **620**. The foreign materials B may affect a quality of the cutting of the metal thin-film substrate **640** to be performed later (e.g., a subsequent cutting), and thus, may need to be removed.

[0106] FIG. **7** illustrates a side view showing a state in which air A is supplied in a punching device according to an embodiment of the present disclosure.

[0107] FIG. **7** may illustrate a state after a punch **720** is lowered to cut a metal thin-film substrate **740**. In some embodiments, FIG. **7** illustrates that the metal thin-film substrate **740** is fixed by a die **730** and a stripper **710**, but the present disclosure is not limited thereto. For example, the process of supplying the air A may be performed in a case where a cut substrate (e.g., the cut substrate **642** of FIG. **6**) is disposed on the die **730** after a first mold is raised and lowered. In other embodiments, the process of supplying the air A may be performed after the cut substrate is moved. In other embodiments, the process of supplying the air A may be performed regardless of whether the first mold is raised or lowered.

[0108] The stripper **710** may include an air supply passage **714**. The air supply passage **714** may supply the air A to a side surface of a punch accommodation portion. In some embodiments, the air A may be supplied through an air supply nozzle **716** formed on the side surface of the punch accommodation portion. In some embodiments, the side surface of the punch **720** may be disposed to face the side surface of the punch accommodation portion in which the air supply nozzle **716** is formed. For example, the air supply nozzle **716** may be disposed to face the side surface of the punch **720**. In this case, air with a stronger pressure may be supplied to the side surface of the punch **720** where a relatively larger amount of powder may be attached.

[0109] Foreign materials B may be attached to the surface of the lifted punch **720**. The air A supplied through the air supply passage **714** may be supplied at a pressure above a suitable level (e.g., a certain or predetermined level) so as to remove the attached foreign materials B. For example, the air A may be supplied at a pressure of 0.1 Mpa or more. The air A may include various suitable components. For example, the air A may be air in the atmosphere. In some embodiments, the air A may be an inert gas.

[0110] In an embodiment, the air supply nozzle **716** may be positioned lower than a punching oil supply nozzle **712** positioned inside the stripper. The punching oil may be supplied from the punching oil supply nozzle **712**, and may flow toward the air supply nozzle **716**. In some embodiments, the air supply nozzle **716** supplies air to the side surface of the punch accommodation portion, thereby preventing or substantially preventing the flowing punching oil from flowing into the air supply nozzle **716**.

[0111] As the punch **720** repeatedly cuts the metal thin-film substrate, the punch **720** may break or cracks may occur. The punching device according to some embodiments of the present disclosure may prevent or substantially prevent damage to the mold by supplying the punching oil. In some embodiments, damage to the punch **720** may be prevented or substantially prevented by supplying the air A to remove the foreign materials B that may be stuck/attached to the surface of the punch **720**. In some embodiments, because no burrs may be formed in the mold, the metal thin-film substrate **740** may be cut more uniformly.

[0112] It may be easier and more stable to supply the punching oil from inside the stripper **710**. For example, in a case where spraying and applying the punching oil to the metal thin-film substrate **740**, the punching oil may not be applied uniformly or substantially uniformly, or a large amount of punching oil may be applied. In other embodiments, the punching device may supply the punching oil directly to the cut region, such that the punching oil may be supplied more evenly.

[0113] In other embodiments, in the case where the punching oil is supplied from inside the punch **720**, the punching oil supply nozzle may be formed on the surface of the punch **720**. Because the punch **720** may be easily damaged by cutting by being in direct contact with the metal thin-film substrate **740**, the punching oil supply nozzle formed on the surface of the punch **720** may be damaged. In other embodiments, because the stripper **710** may not directly cut the metal thin-film substrate **740**, the stripper **710** may not be easily damaged even in a case where the stripper **710** includes the punching oil supply passage, the punching oil supply nozzle **712**, the air supply passage **714**, and/or the air supply nozzle **716**. In this manner, the durability of the punching device according to some embodiments of the present disclosure may be improved.
[0114] FIG. **8** illustrates a perspective view of a stripper **800** according to an embodiment of the present disclosure.
[0115] The stripper **800** may include a plurality of punch accommodation portions **810**, **820**, and **830**. FIG. **8** illustrates three punch accommodation portions **810**, **820**, and **830**, but the present disclosure is not limited thereto. For example, the stripper **800** may include fewer than three or more than three punch accommodation portions.
[0116] In an embodiment, the punch accommodation portions **810**, **820**, and **830** may have the shape of a through-hole. In the assembled punching device, a punch may be inserted through the stripper **800**. In some embodiments, the punch may be inserted into and accommodated in the punch accommodation portions **810**, **820**, and **830**.
[0117] In an embodiment, one or more punching oil supply nozzles **812_1** to **812_4** may be formed on the side surface of each of the plurality of punch accommodation portions **810**, **820**, and **830**. Referring to FIG. **8**, the plurality of punching oil supply nozzles **812_1** to **812_4** may be formed on one side surface of the first punch accommodation portion **810**. However, the present disclosure is not limited thereto, and one or more punching oil supply nozzles may be formed on another side surface of the first punch accommodation portion **810**. For example, one or more punching oil supply nozzles may be formed on each of the plurality of sides of the first punch accommodation portion **810**. One or more punching oil supply nozzles may be formed on each of the first and second side surfaces of the first punch accommodation portion **810**. In this structure, the side surface of the punch may be arranged to be surrounded (e.g., around a periphery thereof) by the first side surface and the second side surface. By forming the punching oil supply nozzles on the plurality of side surfaces of the stripper **800**, the punching oil may be supplied more smoothly.
[0118] The stripper may include a plurality of punching oil supply passages **814_1** to **814_4**. The plurality of punching oil supply nozzles **812_1** to **812_4** may be connected to the plurality of punching oil supply passages **814_1** to **814_4**, respectively. For example, the first punching oil supply passage **814_1** may be connected to the first punching oil supply nozzle **812_1**, such that the punching oil may be supplied from the first punching oil supply passage **814_1** to the first punching oil supply nozzle **812_1**.
[0119] The plurality of punching oil supply passages **814_1** to **814_4** may be connected to one punching oil passage **816**. In some embodiments, the punching oil passage **816** may be branched to form the plurality of punching oil supply passages **814_1** to **814_4**. The plurality of punching oil supply passages **814_1** to **814_4** may be connected to one or more punching oil supply nozzles **812_1** to **812_4** formed in the plurality of punch accommodation portions **810**, **820**, and **830**. Accordingly, the punching oil passage **816** may be connected to the plurality of punching oil supply nozzles **812_1** to **812_4**.
[0120] The punching oil passage **816** may be connected to a pump (e.g., the pump **350** in FIG. **3**). The pump may supply the punching oil to the punching oil passage **816**.
[0121] Thereafter, the punching oil may be supplied to the plurality of punching oil supply passages **814_1** to **814_4** through the punching oil passage **816**. The supplied punching oil may be supplied again to the punching oil supply nozzles **812_1** to **812_4** respectively connected to the plurality of punching oil supply passages **814_1** to **814_4**. In some embodiments, the punching oil

may be sequentially supplied from the punching oil passage **816** to the punching oil supply nozzles **812_1** to **812_4** through the punching oil supply passages **814** to **814_4**.

[0122] FIG. **8** illustrates that the stripper **800** includes one punching oil passage **816**, but the present disclosure is not limited thereto. For example, the stripper **800** may include a plurality of punching oil passages, and a pump may be connected to each of the plurality of punching oil passages. In some embodiments, each of the plurality of punching oil passages **816** may be connected to a separate punching oil supply passage.

[0123] Although an air supply nozzle, an air supply passage, and an air passage are not illustrated, the stripper **800** is not limited thereto. For example, similar to the punching oil supply nozzles **812_1** to **812_4**, an air supply nozzle may be formed inside the stripper **800**, except that air is supplied instead of the punching oil. The air supply passage may be formed similarly to the punching oil supply passages **814_1** to **814_4**, except that air is supplied instead of the punching oil. The air passage may be formed similarly to the punching oil passage **816**, except that air is supplied instead of the punching oil.

[0124] The configuration of supplying the punching oil from inside the punch may be cumbersome because the punching oil supply passage may be formed for each punch, and the punching oil supply passage may be separately controlled for each punch. In other embodiments, the punching device according to some embodiments of the present disclosure may collectively control the supply of punching oil through the punching oil passage **816**, making it easier to manage the supply of the punching oil. In more detail, in the structure in which the plurality of punches are raised and lowered, the punching oil may be effectively supplied to the plurality of punches by forming a passage with a more simple structure only in the stripper **800** without forming the passage in each punch.

[0125] FIG. **9** illustrates a perspective view of a stripper **900** according to an embodiment of the present disclosure.

[0126] The punching oil supply nozzle, the punching oil supply passage, and the punching oil passage formed in the stripper **900** may be understood based on the description provided above with reference to FIG. **8**. With reference to FIG. **9**, the differences from the stripper **800** described above with reference to FIG. **8** may be described in more detail hereinafter.

[0127] In an embodiment, the stripper **900** may include a plurality of punch accommodation portions **910**, **920**, and **930**. The punch accommodation portions **910**, **920**, and **930** may have a structure in which one side surface is opened. Referring to FIG. **9**, at least a portion of the side surface of the stripper **900** may be opened, and thus, one side surface of each of the punch accommodation portions **910**, **920**, and **930** may be opened. In the assembled punching device including the stripper **900**, at least a portion of the side surface of the punch may be surrounded (e.g., around a periphery thereof) by the side surfaces of the punch accommodation portions **910**, **920**, and **930**. In some embodiments, a remaining portion of the side surface of the punch may not face the side surfaces of the punch accommodation portions **910**, **920**, and **930**.

[0128] FIG. **10** illustrates a flowchart showing an example of a punching method **1000** according to an embodiment of the present disclosure. In an embodiment, the punching method S**1000** may be performed by a punching device (e.g., the punching device **100** of FIG. **1**) that includes a die, a stripper, and a punch.

[0129] The punching method S**1000** may start, and a substrate may be supplied between the stripper and the die (S**1010**). In an embodiment, the punching device may supply a punching oil to a side surface of a punch accommodation portion formed in the stripper through a punching oil supply passage formed inside the stripper (S**1020**). The punching device may supply the punching oil through a plurality of punching oil supply passages by using a pump, and may sequentially supply the punching oil through a plurality of nozzles formed on the side surface of the punch accommodation portion.

[0130] In an embodiment, the punching device may supply the punching oil to the side surface of

the punch accommodation portion prior to cutting the substrate. In some embodiments, the punching device may supply the punching oil to the side surface of the punch accommodation portion at the same or substantially the same time as the cutting of the substrate.
[0131] In an embodiment, the punching device may move the punch and the stripper toward the substrate, such that the stripper fixes the substrate and the punch cuts the substrate (S**1030**), and the method S**1000** may end. The side surface of the punch may be disposed to face the side surface of the stripper.
[0132] In an embodiment, the punching device may remove foreign materials attached to the surface of the punch by supplying air to the surface of the punch by using at least one of an air supply passage formed in the stripper or an air supply hole formed in the punch. The punching device may remove the foreign materials attached to the surface of the punch by supplying air to the surface of the punch after punching the substrate.
[0133] However, the present disclosure is not limited to the flowchart illustrated in FIG. **10** and the related description above. For example, one or more processes in the flowchart and the related description may be added, changed, or deleted, the order of one or more of the processes may be variously modified as needed or desired, and/or one or more processes may be performed concurrently or substantially simultaneously with each other.
[0134] The electronic or electric devices and/or any other relevant devices or components according to embodiments of the present disclosure described herein (e.g., the controller and the like) may be implemented utilizing any suitable hardware, firmware (e.g. an application-specific integrated circuit), software, or a combination of software, firmware, and hardware. For example, the various components of these devices may be formed on one integrated circuit (IC) chip or on separate IC chips. Further, the various components of these devices may be implemented on a flexible printed circuit film, a tape carrier package (TCP), a printed circuit board (PCB), or formed on one substrate. Further, the various components of these devices may be a process or thread, running on one or more processors, in one or more computing devices, executing computer program instructions and interacting with other system components for performing the various functionalities described herein. The computer program instructions are stored in a memory which may be implemented in a computing device using a standard memory device, such as, for example, a random access memory (RAM). The computer program instructions may also be stored in other non-transitory computer readable media such as, for example, a CD-ROM, flash drive, or the like. Also, a person of skill in the art should recognize that the functionality of various computing devices may be combined or integrated into a single computing device, or the functionality of a particular computing device may be distributed across one or more other computing devices without departing from the spirit and scope of the example embodiments of the present disclosure.
[0135] Although the present disclosure has been described with reference to embodiments and drawings illustrating aspects thereof, the present disclosure is not limited thereto. Various modifications and variations can be made by a person skilled in the art to which the present disclosure belongs within the scope of the technical spirit of the present disclosure and the claims and their equivalents, below.

## Claims

**1**. A punching device comprising: a die configured to receive a substrate disposed thereon; a stripper spaced apart from and facing the die, and configured to move toward the substrate to fix the substrate; and a punch configured to move toward the substrate and cut the substrate, wherein the stripper comprises: a punch accommodation portion configured to accommodate the punch; and a punching oil supply passage located inside the stripper, and configured to supply a punching oil to a side surface of the punch accommodation portion.
**2**. The punching device as claimed in claim 1, wherein the punch accommodation portion has a

shape of a through-hole through which the punch is configured to be inserted via the stripper.
**3**. The punching device as claimed in claim 1, wherein one side surface of the punch accommodation portion is opened.
**4**. The punching device as claimed in claim 1, wherein the punch accommodation portion comprises a plurality of side surfaces, a nozzle of the punching oil supply passage is located on a first side surface from among the plurality of side surfaces of the punch accommodation portion, and a side surface of the punch is configured to face the first side surface.
**5**. The punching device as claimed in claim 4, wherein the punching oil supply passage comprises a plurality of punching oil supply passages, a first nozzle of a first punching oil supply passage and a second nozzle of a second punching oil supply passage from among the plurality of punching oil supply passages are located on the first side surface of the punch accommodation portion, and the side surface of the punch is configured to face the first side surface.
**6**. The punching device as claimed in claim 4, wherein the punching oil supply passage comprises a plurality of punching oil supply passages, a first nozzle of a first punching oil supply passage from among the plurality of punching oil supply passages is located on the first side surface of the punch accommodation portion, a second nozzle of a second punching oil supply passage from among the plurality of punching oil supply passages is located on a second side surface from among the plurality of side surfaces of the punch accommodation portion, and the side surface of the punch is configured to face the first side surface.
**7**. The punching device as claimed in claim 1, wherein the stripper further comprises an air supply passage located inside the stripper, and configured to supply air to the side surface of the punch accommodation portion.
**8**. The punching device as claimed in claim 1, wherein the punch comprises an air supply hole extending through the punch, and configured to supply air to a bottom surface of the punch.
**9**. The punching device as claimed in claim 7, wherein the air is supplied at a pressure of 0.1 Mpa or more.
**10**. The punching device as claimed in claim 7, wherein the air supply passage is located below the punching oil supply passage.
**11**. The punching device as claimed in claim 1, wherein the stripper comprises a plurality of punch accommodation portions configured to accommodate a plurality of punches, and a plurality of punching oil supply passages configured to supply the punching oil, a first nozzle of a first punching oil supply passage from among the plurality of punching oil supply passages is located on a side surface of a first punch accommodation portion from among the plurality of punch accommodation portions, a second nozzle of a second punching oil supply passage from among the plurality of punching oil supply passages is located on a side surface of a second punch accommodation portion from among the plurality of punch accommodation portions, a side surface of a first punch is configured to face the side surface of the first punch accommodation portion, and a side surface of a second punch is configured to face the side surface of the second punch accommodation portion.
**12**. The punching device as claimed in claim 1, wherein the punching oil is supplied in a fixed-quantity discharge method.
**13**. The punching device as claimed in claim 1, wherein the punching oil is supplied at a predetermined flow rate.
**14**. The punching device as claimed in claim 1, further comprising a controller configured to adjust an amount of the punching oil, wherein the controller is configured to control a pump connected to the punching oil supply passage to supply the punching oil in proportion to a number of times the punch is raised and lowered.
**15**. The punching device as claimed in claim 7, further comprising a controller configured to adjust an amount of the air, wherein the controller is configured to control an air pump connected to the air supply passage to supply the air in proportion to a number of times the punch is raised and

lowered.

**16**. A punching method comprising: supplying a substrate between a stripper and a die; supplying a punching oil to a side surface of a punch accommodation portion of the stripper through a punching oil supply passage located inside the stripper; and moving a punch and the stripper toward the substrate, such that the stripper fixes the substrate and the punch cuts the substrate, wherein a surface of the punch is disposed to face a side surface of the stripper.

**17**. The punching method as claimed in claim 16, wherein the supplying of the punching oil to the side surface of the punch accommodation portion comprises supplying the punching oil to a plurality of punching oil supply passages by a pump, such that the punching oil is sequentially supplied through a plurality of nozzles located on the side surface of the punch accommodation portion.

**18**. The punching method as claimed in claim 16, further comprising removing foreign materials attached to the surface of the punch by supplying air to the surface of the punch using at least one of an air supply passage located in the stripper or an air supply hole located in the punch.

**19**. The punching method as claimed in claim 16, wherein the supplying of the punching oil to the side surface of the punch accommodation portion comprises supplying the punching oil to the side surface of the punch accommodation portion before cutting the substrate.

**20**. The punching method as claimed in claim 18, wherein the removing of the foreign materials attached to the surface of the punch comprises removing the foreign materials attached to the surface of the punch by supplying air to the surface of the punch after punching the substrate.